Cálculo Diferencial - Actividad 6

Resolver los siguientes ejercicios de forma analítica y comprobar los resultados con Python.

Hallar la derivada de las siguientes funciones:

1. $y = \ln(ax + b)$
2. $y = \ln\left(ax^2 + b\right)$
3. $y = \ln(ax + b)^2$
4. $y = \ln(ax^n)$
5. $y = \ln\left(x^3\right)$
6. $y = \ln(x)^3$
7. $y = \ln\left(2x^3 - 3x^2 + 4\right)$
8. $y = \log\left(\frac{2}{x}\right)$
9. $y = \ln\left(\frac{x^2}{1+x^2}\right)$
10. $y = \ln\left(\sqrt{9 - 2x^2}\right)$
11. $y = \ln\left(ax\sqrt{a + x}\right)$
12. $y = x\ln(x)$
13. $y = \ln\left(x + \sqrt{1 + x^2}\right)$
14. $s = \ln\left(\sqrt{\frac{a+bt}{a-bt}}\right)$
15. $y = x^2\ln\left(x^2\right)$
16. $y = e^{nx}$
17. $y = 10^{nx}$
18. $y = e^{x^2}$
19. $y = \frac{2}{e^x}$
20. $s = e^{\sqrt{t}}$
21. $z = b^{2y}$
22. $u = se^s$
23. $v = \frac{e^u}{u}$
24. $y = \frac{\ln(x)}{x}$
25. $y = \ln\left(x^2e^x\right)$
26. $y = \frac{e^x-1}{e^x+1}$
27. $y = x^2e^{-x}$
28. $y = \frac{e^x-e^{-x}}{e^x+e^{-x}}$
29. $s = \frac{\ln\left(t^2\right)}{t^2}$
30. $y = \ln\left(\frac{\sqrt{x^2+1}-x}{\sqrt{x^2+1}+x}\right)$

En los problemas 31 a 40 hallar el valor de $\frac{\mathrm{d}y}{\mathrm{d}x}$ para el valor dado de x.

31. $y = \ln\left(x^2 + 2\right);\ x = 4$
32. $y = \log(4x - 3);\ x = 2$
33. $y = x\ln\left(\sqrt{x + 3}\right);\ x = 6$
34. $y = xe^{-2x};\ x = \frac{1}{2}$
35. $y = \frac{\ln\left(x^2\right)}{x};\ x = 4$
36. $y = \frac{e^{\frac{x}{2}}}{x+1};\ x = 1$
37. $y = \log\sqrt{25 - 4x};\ x = 5$
38. $y = 10^{\sqrt{x}};\ x = 4$
39. $y = \left(\frac{3}{x}\right)^x;\ x = 3$
40. $y = \frac{x^3\sqrt{x^2+9}}{\sqrt[3]{20-3x}};\ x = 4$